

v17.21 Completion Report — FINAL EQ Line Fix + HUD Text Uniformity

Date: 2026-05-16
Branch: feat/v17.19-eq-fix-hud-visibility
Commit: feat: v17.21 FINAL EQ line fix + HUD text uniformity
Type: Visual-Only Update (ZERO logic/filter/alert changes)
Plot Budget: 33/64 (unchanged)

Delivery Checklist

Phase 1: Backup & Setup

- [x] Created backup: smc_premium_discount_indicator_v17.20_backup.pine
- [x] Created working copy → finalized as smc_premium_discount_indicator_v17.21.pine

Phase 2: FIX 1 — HUD Text Uniformity

- [x] Found MSS-Retrace color: color.new(color.silver, 0) (used for Entry Mode row)
- [x] Applied uniform silver color to ALL 16 value cells (right column)
- [x] REMOVED all conditional dimming (color.new(color.gray, 30)) for inactive rows
- [x] Color exceptions preserved:
- INVALIDATED / killed → color.new(color.red, 0) ✓
- TRAP BUY / TRAP SELL → color.new(color.orange, 0) ✓
- ALIGNED → color.new(color.lime, 0) ✓
- Signal fired (FIRE stage, alert fired, signals today > 0) → color.new(color.lime, 0) ✓
- Everything else → color.new(color.silver, 0) ✓

Rows Updated:

Row	Label	Old Color Logic	New Color
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1	Bias	red/green/gray (zone-based)	silver (uniform)
2	Align	red/lime (conflict/aligned)	KEPT — red/lime exceptions
3	KZ	green/gray30	silver (uniform)
4	Sniper	green/yellow/gray30	silver (uniform)
5	POI	green/yellow/gray30	silver (uniform)
6	Momentum	orange/gray30	silver (uniform)
7	Guardian	green/yellow/gray30	orange (TRAP) / silver (all else)
8	Confl	green/yellow/orange/gray30	silver (uniform)
9	AMD	orange/green/gray30	silver (uniform)
10	Trade	green/red/yellow/gray30	silver (uniform)
11	Entry	silver (was already correct)	silver (unchanged)
12	Action	red/purple/green/orange/blue/gray	red (INVALIDATED) / silver (all else)
13	Stage	green/lime/yellow/orange/dim	lime (FIRE) / silver (all else)
14	Alert	green/red/gray30	lime (fired) / silver (none)

| 15 | Signals | green/dim | lime (>0) / silver (0) |

| 16 | Levels | aqua/dim | silver (uniform) |

Phase 3: FIX 2 — EQ Line FINAL ROOT CAUSE FIX

Root Cause Identified: Previous versions used `barstate.islast` guard, meaning the EQ line was only drawn on the very last bar. If zone boxes rendered after, or if the chart reloaded, the line could be painted over.

EXACT Implementation (User's Specified Pattern):

```
// Runs on EVERY bar (no barstate.islast guard)
if showH4Zones and showEquilibriumLine
    float eq_level = (h4RangeHigh + h4RangeLow) / 2.0

    if not na(pdEqLine)
        line.delete(pdEqLine)
    pdEqLine := line.new(
        x1 = bar_index - 1,
        y1 = eq_level,
        x2 = bar_index,
        y2 = eq_level,
        color = color.white,
        width = 5,
        style = line.style_solid,
        extend = extend.right
    )

    if not na(pdEqLabel)
        label.delete(pdEqLabel)
    pdEqLabel := label.new(
        x = bar_index,
        y = eq_level,
        text = "— EQ",
        color = color.new(color.black, 0),
        textcolor = color.white,
        style = label.style_label_left,
        size = size.normal
    )
```

Key Changes from v17.20:

- [x] REMOVED `barstate.islast` guard — now runs on every bar
- [x] `x1 = bar_index - 1, x2 = bar_index` (1-bar segment with `extend.right`)
- [x] `extend = extend.right` for full rightward chart coverage
- [x] `eq_level = (h4RangeHigh + h4RangeLow) / 2.0` — exact 50% midpoint
- [x] Width = 5, color = white, style = solid
- [x] Label uses same delete-and-redraw pattern
- [x] Label text = "— EQ" with solid black background

Phase 4: Version Updates

- [x] Pine Script title: v17.20 → v17.21
- [x] HUD header cell: v17.20 → v17.21
- [x] All 4 alert JSON payloads: v17.20 → v17.21
- [x] Changelog header added for v17.21

Phase 5: Backend Sync (16 files updated)

- [x] `app.py` — logging version
- [x] `src/webhook_server/routes.py` — sample payloads + comments
- [x] `src/oie_processor.py` — version detection
- [x] `src/decoders.py` — internal version string
- [x] `src/database.py` — migration logging
- [x] `src/oie_database.py` — default version
- [x] `templates/base.html` — title + footer badge
- [x] `templates/settings.html` — headers + alert templates
- [x] `templates/dashboard.html` — page title
- [x] `templates/trades.html` — page title
- [x] `templates/opportunities.html` — page title + badge
- [x] `schemas/migrate_v17_14_oie.sql` — comment + default version
- [x] `tradex-frontend/app/layout.tsx` — title + header + footer
- [x] `tradex-frontend/app/page.tsx` — content + badge + decode ref
- [x] `tradex-frontend/app/opportunities/page.tsx` — subtitle + placeholder
- [x] `tradex-frontend/app/settings/page.tsx` — header + templates + version

Phase 6: Git

- [x] Committed with descriptive message
 - [x] Pushed to `feat/v17.19-eq-fix-hud-visibility` branch
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Modified Files (20 total)

#	File	Change Type
1	smc_premium_discount_indicator_v17.21.pine	NEW — main indicator
2	smc_premium_discount_indicator_v17.20_backup.pine	NEW — backup
3	app.py	Version string
4	src/webhook_server/routes.py	Version strings
5	src/oie_processor.py	Version detection
6	src/decoders.py	Version string
7	src/database.py	Version strings
8	src/oie_database.py	Default version
9	templates/base.html	Title + footer
10	templates/settings.html	Templates + headers
11	templates/dashboard.html	Page title
12	templates/trades.html	Page title
13	templates/opportunities.html	Title + badge
14	schemas/migrate_v17_14_oie.sql	Comment + default
15	tradex-frontend/app/layout.tsx	Title + badges
16	tradex-frontend/app/page.tsx	Content + badges
17	tradex-frontend/app/opportunities/page.tsx	Subtitle
18	tradex-frontend/app/settings/page.tsx	Templates
19	v17.21_completion_report.md	This report

TradingView Installation Steps

1. Open TradingView → Pine Editor
2. Remove the existing v17.20 indicator from your chart
3. Paste the full contents of `smc_premium_discount_indicator_v17.21.pine`
4. Click “Add to Chart”
5. Verify:
 - HUD header shows **“SMC P&D v17.21”**
 - All HUD value text is uniform bright silver
 - Only exceptions show color: INVALIDATED (red), TRAP (orange), ALIGNED (lime), signals (lime)
 - **EQ line is visible** as a thick white line at the 50% midpoint
 - EQ label shows “— EQ” with black background

ZERO Logic Changes Verification

- ☒ No changes to confluence scoring (8 components)
- ☒ No changes to Guardian tier logic (SNIPER/RETRACE/TRAP)
- ☒ No changes to Displacement Trap state machine
- ☒ No changes to alert firing conditions
- ☒ No changes to signal deduplication
- ☒ No changes to MSS-Retrace detection
- ☒ No changes to Kill Zone definitions
- ☒ No changes to H4 Bias calculation
- ☒ Plot count unchanged: 33/64